INTERNSHIP REPORT • 2026

Siemens Data Science Master Virtual Internship

An eight-week learning journey across data engineering, machine learning, data preparation, AI applications, and enterprise platform administration.

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Roll No. 2410030778 • Batch 3CSE12 (2024–2028)

APRIL – JUNE 2026

AICTE–EduSkills Virtual Internship Program | Supported by Siemens

Internship at a glance

A structured, assessment-driven virtual program designed to connect data science concepts to an end-to-end workflow.

08

WEEKS

Structured learning journey

APR–JUN

2026

Internship period

O

OUTSTANDING

Internship grade

AI HUB

PLATFORM

RapidMiner AI Hub exposure

PROGRAM MODEL

Weekly modules → Weekly assessments → Assignments / documentation where applicable → Final Assessment Test

Organized by the AICTE–EduSkills Virtual Internship Program, with the Siemens Data Science Master track and a progressive professional-to-master curriculum.

The challenge and the objectives

The internship focused on bridging separate academic topics into one coherent, practical data science lifecycle.

PROBLEM STATEMENT

Turning raw data into useful decisions

Organizations generate large amounts of data. The challenge is converting it into actionable insights and connecting data access, preparation, modelling, deployment, and monitoring instead of treating them as isolated topics.

CORE GAP

From individual concepts to an end-to-end workflow

OBJECTIVES

- Identify business problems suited to AI and Machine Learning.
- Understand data access, loading, transformation, and preparation.
- Build foundational and advanced Machine Learning knowledge.
- Study model deployment, monitoring, parameter optimization, and time-series forecasting.
- Gain working knowledge of RapidMiner AI Hub administration and enterprise workflows.
- Complete weekly assessments and final credential validation.

Five learning pillars

The week-wise curriculum moved from professional-level foundations to advanced master-level modules.

01 Applications & Use Cases

Identify business problems where AI and Machine Learning can create value.

02 Data Engineering

Access, load, transform, and calculate data for analysis and modelling.

03 Machine Learning

Build foundational and advanced knowledge of models and workflows.

04 Data Preparation

Clean, transform, organize, and automate data handling.

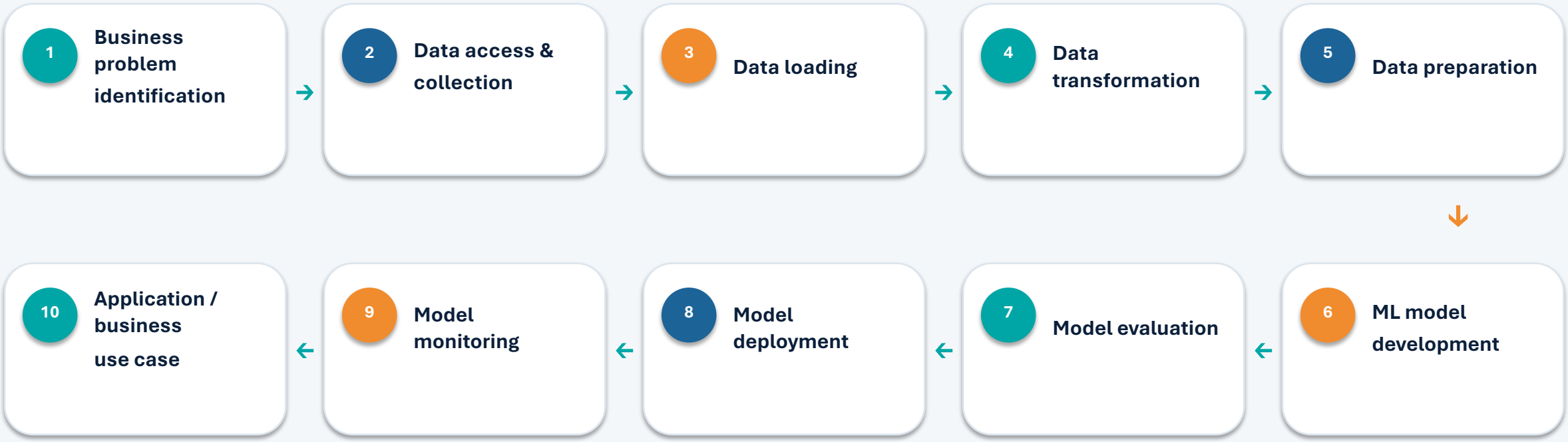
05 Platform Administration

Configure and manage RapidMiner AI Hub and enterprise AI infrastructure.

Additional master-level focus: practical model deployment and monitoring, advanced Machine Learning, and platform operations.

End-to-end Data Science workflow

A pipeline view of the stages covered across the internship modules.



Problem → Data → Preparation → Model → Application → Deployment → Monitoring

Methodology: how the learning progressed

A structured weekly format reinforced concepts before moving into more advanced modules.

01

Conceptual learning

Week-wise topics introduced in a logical sequence.

02

Practical exposure

Apply concepts through platform-oriented activities.

03

Weekly assessment

Check understanding and reinforce module learning.

04

Assignments & documentation

Complete submissions where applicable.

05

Professional → Master

Progress from foundations to advanced modules.

06

Final credential validation

Final Assessment Test and grade point assessment.

Outcome: a measured progression from core concepts to deployment and enterprise platform administration.

Week-by-week journey | Weeks 1–4

The first half established business context, data handling, Machine Learning foundations, and advanced preparation.

WEEK 01	**Applications & Use Cases** **Professional**	Identify business problems suitable for AI and Machine Learning; explore real-world applications.
WEEK 02	**Data Engineering** **Professional**	Learn to access, load, transform, and calculate data for analysis and modelling.
WEEK 03	**Machine Learning** **Professional**	Develop foundational understanding of core models, workflows, and evaluation methods.
WEEK 04	**Data Preparation** **Master**	Advance data handling, transformation, and automation techniques.

Each module was reinforced through weekly assessments as part of the overall completion requirements.

Week-by-week journey | Weeks 5–8

The second half shifted toward advanced Machine Learning, operational use, and platform administration.

WEEK 05	**Machine Learning Master**	Advanced ML techniques and how data preparation choices can influence model performance.
WEEK 06	**Applications & Use Cases Master**	Practical model deployment and monitoring approaches for production use cases.
WEEK 07	**Platform Administration Master**	Configuration and management, with exposure to RapidMiner AI Hub.
WEEK 08	**Platform Administration Master**	Enterprise platform operations, administration, and reliable AI infrastructure.

FINAL ASSESSMENT

Completed weekly assessments, applicable assignments, and the Final Assessment Test for credential validation and grade assessment.

Technology areas and platform exposure

The modules connected data operations, model development, deployment, monitoring, and enterprise platform familiarity.

DATA OPERATIONS

- Data access and loading
- Transformation and calculation
- Cleaning and preparation
- Workflow-ready datasets

MODEL DEVELOPMENT

- Foundational ML models
- Advanced ML concepts
- Parameter optimization
- Time-series forecasting

DEPLOYMENT & PLATFORM

- Model deployment and monitoring
- RapidMiner AI Hub
- Platform configuration
- Enterprise operations

RapidMiner AI Hub was the named enterprise platform in the internship report.

Learning outcomes and skills developed

The internship strengthened both technical understanding and the habits needed for structured industry-style work.

TECHNICAL OUTCOMES

- Understand the end-to-end Data Science workflow.
- Access, load, transform, and prepare data for analysis.
- Build foundational and advanced Machine Learning knowledge.
- Understand AI applications and real-world use cases.
- Gain exposure to model deployment, monitoring, and platform administration.

PROFESSIONAL SKILLS

- Analytical thinking and problem-solving.
- Self-learning discipline and consistency.
- Time management across weekly modules.
- Documentation and assessment readiness.
- A clearer understanding of how data science components fit together in practice.

Challenges and learning experience

The biggest learning shift was moving from isolated topics toward one connected operational workflow.

1

THE CHALLENGE

Connecting data engineering, preparation, modelling, deployment, monitoring, and administration into one lifecycle.

2

WHAT HELPED

The week-wise structure built each module on the previous one; weekly assessments reinforced concepts before progressing.

3

KEY TAKEAWAY

Moving from professional to master-level content required more effort, particularly in advanced ML and platform administration.

The program structure helped turn a broad field into a step-by-step, assessable learning path.

Conclusion and future scope

The internship provided a practical foundation for continued learning in Data Science, Machine Learning, and AI applications.

NEXT LEARNING DIRECTIONS

Deep Learning

Move into deeper neural-network approaches.

Real-world datasets

Work with larger and more varied datasets.

Cloud data engineering

Explore cloud-based data pipelines.

MLOps

Learn deployment and monitoring at scale.

Big Data

Explore technologies for large-scale data.

OVERALL TAKEAWAY

A clearer view of how Data Science, AI, and Machine Learning come together—from problem definition through deployment and monitoring.

Internship offer letter

The letter confirms selection for the 8-week AICTE–EduSkills Virtual Internship Program in the Siemens Data Science Master domain.

REFERENCE NUMBER

C16Y2026M091397690

PROGRAM PERIOD

April 2026 – June 2026

COMPLETION REQUIREMENTS

Weekly assessments, applicable assignments / documentation, and Final Assessment Test



EduSkills
Nation Building Through Skills

Ref No: C16Y2026M091397690

Internship Offer Letter

Student Name	: Kartikeya Khare
Stream	: B.Tech
Branch	: B.Tech(Computer Science and Engineering)
Institute Name	: IILM University, Greater Noida
Internship Domain	: Siemens Data Science Master
Internship Duration	: April 2026 - June 2026
AICTE Student ID	: STU6998573a168e01771591482

We are pleased to inform you that you have been selected for the 8-Week **AICTE – EduSkills Virtual Internship Program**.

During this internship, you will learn and demonstrate industry-relevant skills that will enhance your employability and strengthen your confidence in the subject area.

The program is designed to provide structured learning combined with practical exposure.

Internship Structure & Evaluation:

- The internship will follow a **week-wise structured learning plan**.
- You will be required to complete **weekly assessments** to track your progress and understanding.
- Certain modules will require submission of **project documentation and assignments**.
- In the final week, you must appear for a **Final Assessment Test**.
- The final grade will be based on your overall performance, including weekly assessments, project submissions, and the final test.

Upon successful completion of all required steps and assessments, you will be awarded a **Virtual Internship Certificate**.

Please note that this offer letter will be considered valid only after the successful completion of the internship program and fulfillment of all assessment requirements, including obtaining the final completion certificate. Without the final completion certificate, this offer letter shall not be treated as a valid document for any official purpose.

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COMPLETION PROOF

Certificate of Virtual Internship

Successfully completed the 8-week Data Science Master Virtual Internship, supported by Siemens, during April–June 2026.

INTERNSHIP GRADE	O — Outstanding
CERTIFICATE ID	469bb51ab4e288ddd94a
STUDENT ID	STU6998573a168e01771591482
INSTITUTION	IILM University, Greater Noida

Thank you

